

Supplementary Information

JobMatch: An Agentic Multi-Role Platform for Explainable Job–Candidate Matching

Scope

This supplementary note lists benchmark artifacts and regeneration commands referenced in the main manuscript. It does not introduce new experimental results.

Primary benchmark artifacts

Artifact	Path (repository root)
Paper progression summary	data/expected/paper_progression_summary.json
Fusion ablation	backend/benchmark_outputs/table11_fusion.json
Composite ablation	docs/research/evaluation/artifacts/runs/2026-05-27T150000Z/ablation/
Method comparison	docs/research/evaluation/artifacts/runs/2026-05-27T150000Z/compariso
Cross-encoder report	docs/research/evaluation/artifacts/runs/2026-05-27T150000Z/cross_en
Hard negatives	backend/benchmark_outputs/hard_negatives.json
Fairness audit table	docs/research/evaluation/paper_tables/table4_fairness.csv
Eval corpus	data/cvs.json, data/jobs.json, data/eval_pairs.json

Regeneration commands

```
python backend/scripts/run_research_pipeline.py
python -m benchmarks.paper_progression
python -m pytest tests/benchmarks/test_eval_regression.py -q
```

Regression gate

Committed progression baselines are gated by `tests/benchmarks/test_eval_regression.py` with ± 0.04 nDCG tolerance on nDCG@5.

Aligned with manuscript declarations (May 2026).